# Sizing rooftop PV and batteries after energy efficiency: an optimisation framework for manufacturing SMEs

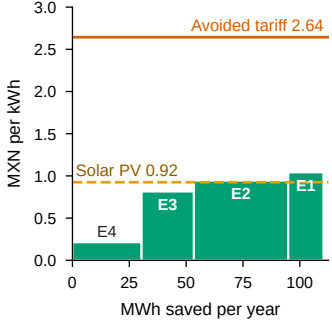
**Data a typical SME already has**

- 28 monthly bills
- Production records
- Walk-through audit
- Official tariffs, emission factor, fuels
- NASA POWER weather

Bills reproduced within 2.2%

## 1 Cut waste

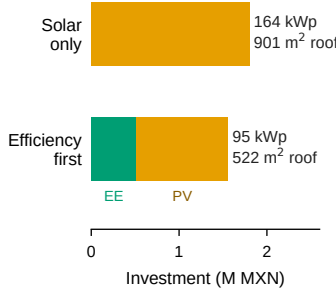
Cost of saved electricity




**−23% grid electricity for 0.52 M MXN**

## 2 Then size solar

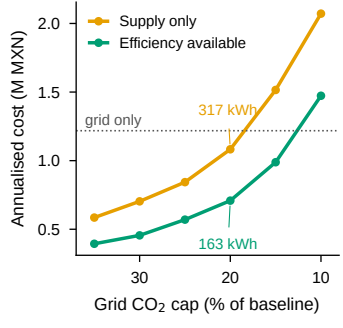
Same 54% cut in grid purchases




**−13% capital, −42% roof, higher NPV in 97% of runs**

## 3 Optimise deep cuts (urbs)

Least-cost system under $CO_2$ caps




**Efficiency enters first: −29–35% cost, about half the storage**

**Net zero, security and resilience**

- Annual net-zero electricity pays: 234 kWp after efficiency vs 303 kWp without
- If PV loses demand credit: solar NPV −69%, efficiency unaffected
- 111 $tCO_2$ and 850 GJ of imported gas avoided per year